

# TITLE PAGE

- Problem Statement ID – Q1
- Problem Statement Title- Eco-Loop Building Agents
- Theme- Sustainability / Physical AI for Smart Buildings
- PS Category- Software
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# ECO-LOOP BUILDING AGENTS

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## Proposed Solution

- **Detailed explanation:**
  - A closed-loop AI agent that reads live EnergyPlus building-simulation state (zone temperature, thermal comfort, live electricity demand) every simulated hour and writes HVAC setpoints back into the running simulation via a tool-calling interface
- **How it addresses the problem?**
  - It replaces rigid, rule-based BMS scheduling with real-time reasoning that adapts to weather, occupancy comfort, and grid carbon intensity, proven end-to-end on a full 31-day simulated run with zero crashes
- **Innovation:**
  - An open-source LLM (Llama 3 8B, self-hosted via Ollama) makes autonomous, auditable HVAC decisions through a guarded tool layer. Every setpoint passes through a hard-coded comfort clamp so the agent can never violate occupant safety bounds

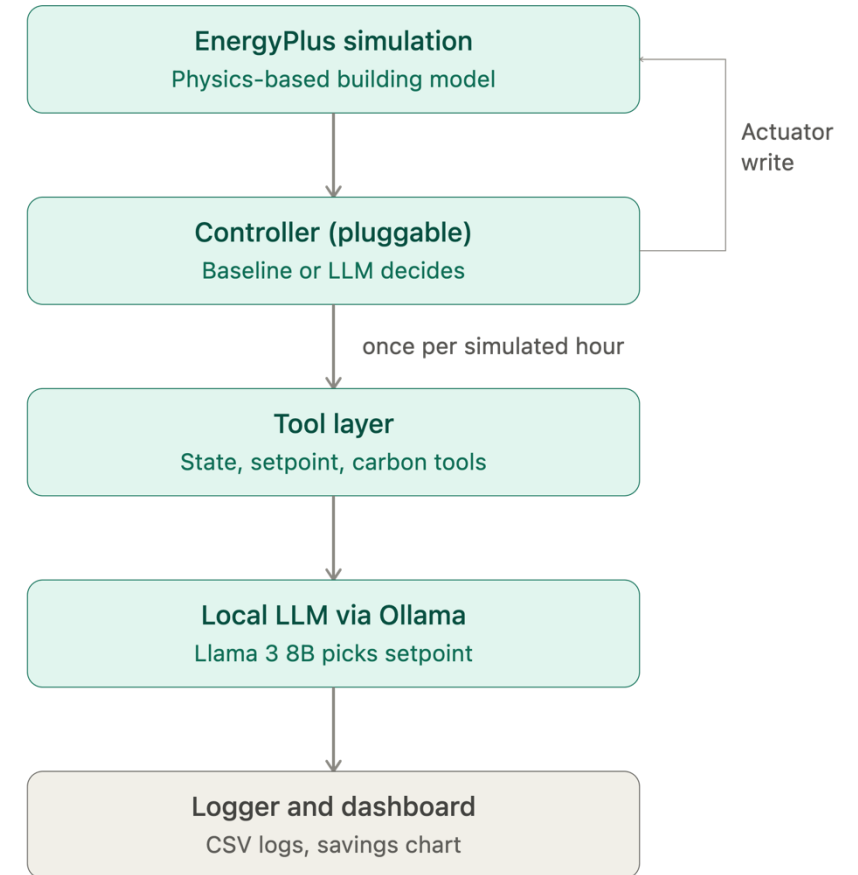
# TECHNICAL APPROACH

## ➤ Technologies used:

Python, EnergyPlus (pyenergyplus API) as the physics-based building simulator, Llama 3 8B via Ollama as the self-hosted open-source LLM, a hand-rolled JSON tool-calling layer (MCP-schema compatible), pandas/matplotlib for the results dashboard

## ➤ Methodology:

EnergyPlus streams live sensor state (zone temp, comfort, electricity, outdoor temp) every simulated hour → three tool functions (get\_building\_state, set\_zone\_setpoint, get\_grid\_carbon\_intensity) format and clamp the data → the LLM reasons over state + a 3-hour rolling history and returns a structured tool call → the clamped setpoint is injected back into the running EnergyPlus instance, closing the loop



# FEASIBILITY AND VIABILITY

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## ➤ **Feasibility:**

Already validated as a working proof-of-concept on a full 31-day (744-hour) closed-loop simulation, ran end-to-end with zero crashes and zero LLM parse failures, achieving a 1.6% energy reduction and improving thermal comfort from 84% to 94% in-band versus a fixed-schedule baseline.

## ➤ **Challenges and risks:**

- LLM inference latency runs in lockstep with the simulation clock (EnergyPlus pauses during each ~4–8s LLM call)
- A self-hosted 8B model can converge on an overly uniform strategy if not given explicit conditional guidance
- Real deployments would need a live building's sensor/BMS integration in place of a simulated one

## ➤ **Strategies for overcoming these challenges:**

- Cap LLM calls at one per simulated hour (not per timestep) to bound latency impact; add explicit prompt guardrails tying setpoint choices to outdoor temperature and time-of-day rather than a single fixed heuristic
- Every setpoint passes through a hard comfort clamp regardless of model output, and any LLM failure falls back to safe fixed setpoints, so the loop never crashes

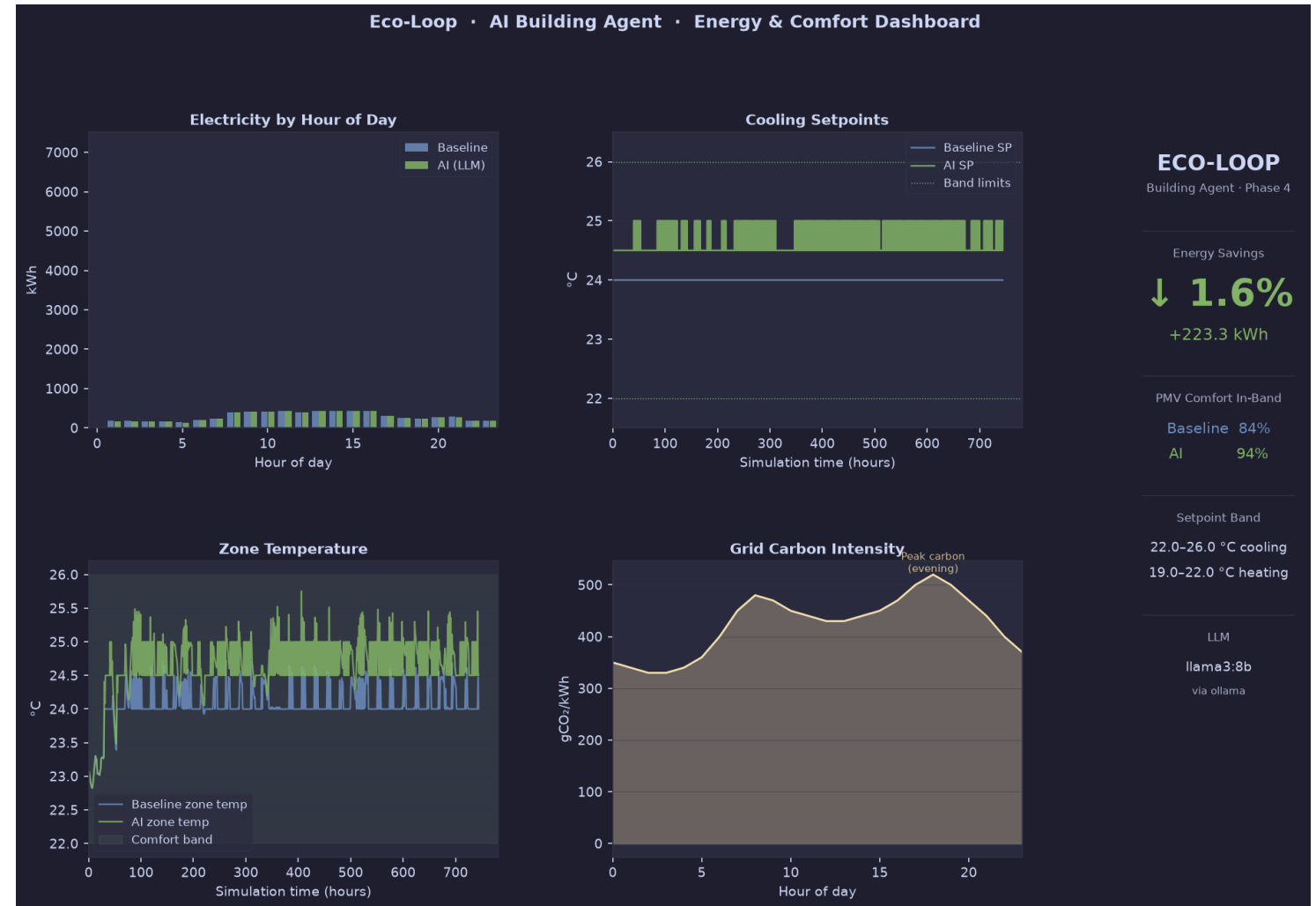
# ARTIFACTS

GitHub repository link (contains full codebase, .idf, build files, architecture.md):

<https://github.com/BuenaPeninnahQuadros/Eco-loop>

Video:

[https://drive.google.com/file/d/1\\_rJzW-qzVXMf45F6NMWTlk3diV08baui/view?usp=sharing](https://drive.google.com/file/d/1_rJzW-qzVXMf45F6NMWTlk3diV08baui/view?usp=sharing)



# RESEARCH AND REFERENCES

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EnergyPlus Python API documentation  
([energyplus.net](https://energyplus.net))

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Model Context Protocol specification  
([modelcontextprotocol.io](https://modelcontextprotocol.io))

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Ollama ([ollama.com](https://ollama.com))

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DOE Commercial Reference Buildings  
(RefBldgSmallOffice)

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ASHRAE 55 thermal comfort / Fanger  
PMV model